

WireMetry

Wi-Fi network monitoring & analytics platform — **multi-vendor**

WireMetry is an on-premise monitoring, troubleshooting and decision-support solution for controller-based (WLC) Wi-Fi infrastructures. It aggregates the live status of access points, clients and the RF environment in real time.

Designed to be **vendor-agnostic**, it supervises **Cisco Catalyst 9800** and **Ruckus SmartZone** controllers from a single web interface, detects security and performance anomalies, and delivers concrete optimization recommendations.

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Multi-vendor

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RF diagnostics

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Wi-Fi security

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Trends

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On-premise

Real-time monitoring

- **Dashboard:** CPU, memory, interfaces, controller version, AP & client counts, WLAN status.
- **Access-point inventory:** model, status, version, channel, band, RF load, clients.
- **Clients:** RSSI, SNR, channel, radio type; search and statistics.
- **Controller connectivity:** connected / degraded / unreachable.

Site mapping

Import **floor plans** and place access points, with overlays:

- Live AP status
- Signal heatmap
- Roaming paths
- Coverage / channels
- Client locator

RF diagnostics

- Channel conflicts: co-channel, adjacent / overlapping interference.
- Airtime utilization and noise level per radio.
- **Spectrum analyzer** per band (2.4 / 5 GHz) with channel width.

Security event log

- Detection: rogue APs, RF events, SNR drops, duplicate addresses, aWIPS alerts, offline APs.
- **Persistence** in database, **acknowledgement** and history.

Decision support

Trends & capacity

Load, clients and RF occupancy over time — saturation forecasting.

Lifecycle & firmware

AP reboot / flap tracking and firmware compliance vs a target version.

Network advisor

Prioritized recommendations: RF conflicts, saturation, offline APs, firmware, weak clients, 2.4 GHz imbalance, controller load.

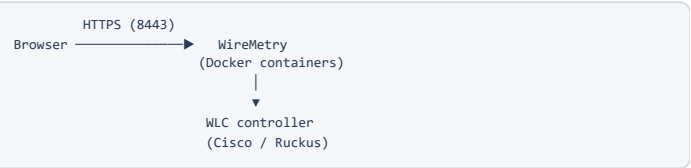
Tracking & roaming

Client connection and roaming history, with **scheduled purge** (5 min / hourly / daily / weekly / monthly) to keep data volume under control.

KEY POINTS

- Cisco Catalyst 9800 & Ruckus SmartZone, single interface
- RF diagnostics: channel conflicts, spectrum, airtime, noise
- Security log: rogues & anomalies, acknowledgeable
- Network advisor: prioritized recommendations
- Docker deployment, HTTPS — your data stays on-site

Architecture & deployment



Deployment	Docker / Docker Compose containers
Access	Secure web interface — HTTPS (port 8443)
Hosting	On-premise — your data stays on-site
WLC integration	Cisco Catalyst 9800 · Ruckus SmartZone

Security

- **Secure-token** authentication, session expiry, admin / viewer roles.
- **Strong password hashing**, forced rotation on first login.
- Brute-force protection (login attempt limiting).
- **Encryption at rest** of controller credentials.
- Authenticated database, not publicly exposed.
- Hardened web security headers (anti-clickjacking, HSTS).
- End-to-end **TLS** encryption; runs as a non-privileged container.

Requirements & sizing

Profile	APs	Clients	vCPU	RAM	Disk
Small	≤ 250	≤ 1,000	2	4 GB	40 GB
Medium	≤ 1,500	≤ 3,000	4	8 GB	80 GB
Large	≤ 6,000	≤ 5,000	4–8	16 GB	150–200 GB

Disk size depends on history retention (scheduled purge).

System	Linux (Ubuntu 22.04+), Docker + Compose
Network	HTTPS/API access to the controller
Client device	Modern browser (Chrome, Edge, Firefox)

Deployment

- **Turnkey** install via Docker Compose.
- Single integrated deployable — **HTTPS access on port 8443**.
- Controller configuration from the interface, **no restart**.